

Course Name: Java Programming

Course Code: DSC-C-CS-352T

Credits: 4

Course Outcomes:

The aim of this course is to enable students to

- **CO1:** Understand the concepts of Object-Oriented Programming Language and using Java
- **CO2:** Get good understanding of developing multi-user applications using Java Programming Language
- **CO3:** Get good understanding of developing multi-threaded applications using Java Programming Language
- **CO4:** Harness the features of Java using APIs of Collection Framework, Lambda expressions and streams for effective programming

Prerequisites:

Basic knowledge of C/C++ programming language

Contents:

Unit No.	Particulars	Credits	Time
1.	Unit Title: Introduction, Data Type and Control Structures <i>Introduction:</i> Recap of Object Oriented Programming Features and Relating to Java, Features of the Java Language, Java Environment, Object Oriented Programming in Java, Java Program Structure, Java and Unicode, Writing First Java Application, Command-Line Arguments, Use of System.out, System.console().printf and System.console().readline() Methods <i>Data Type, Variables and Constants, Loops and Logic & Exceptions:</i> Primitive Types, Reference Types, Difference from C++ in Usage of Reference Types, Arithmetic Calculations, Mixed Arithmetic Expressions, Mathematical Functions and Constants, Bitwise Operators, Enumerated Data Type, Program Comments, Loops, Logic and Scope, Making Decisions, Logical Operators, The Conditional Operator, The Switch Statement <i>Exceptions:</i> Understanding Exceptions, Types of Exceptions, Throw and Throws Keywords, Checked and Unchecked Exceptions, Handling Exceptions using Try-Catch and Finally Blocks, Multi-Catch Blocks, Creating Custom Exceptions	01	15 Hrs.
2.	Unit Title: Defining Classes, Extending classes and Inheritance, Interface Significance of Class, Defining Classes, Various Members for (which go in a) Class Definition – Instance Variable, Class Variable, Methods, Static Methods, Constructors, Initializer Block and Class Initializer Block, Method and Constructor Overloading, Using Objects, Use of <i>this</i> Keyword, Nested Classes, The Finalize() Method, Recap of Inheritance, Use of Keywords Extends and Super, “Is-A” Relationship for Inheritance, Packages and Access Specifiers, Package Keyword, Import and Import Static Keywords, Private, Public And Protected Keywords, Abstract Classes, Abstract Methods, Multiple Inheritance using Interfaces, Static and Default Methods in	01	15 Hrs.

Unit No.	Particulars	Credits	Time
	Interfaces, Functional Interfaces and Syntax of Lambda Expression		
3.	Unit Title: Commonly used Classes, Packages: Custom Packages, In-built Packages Commonly used Classes like String, StringBuilder, Math and Wrapper Classes, Creating and Importing Custom Packages, The java.io Package, File Class, IO Stream Classes for Reading and Writing to Files, RandomAccessFile Class, Stream API from java.util.stream Package, Stream, IntStream, LongStream, DoubleStream, Understanding the Intermediate and Terminal Operations on Streams, <i>reduce</i> and <i>collect</i> methods, Understanding the Collector, Creating useful Collector using methods from the Collectors Class LocalDate, LocalTime, LocalDateTime, OffsetDate, ZonedDateTime, LongSummaryStatistics, from java.util Package, Understanding the Optional Class methods for <i>classification and regression</i> : Bagging, Boosting, Random Forest	01	15 Hrs.
4.	Unit: Threads And Collection Framework Understanding Threads, Thread Class, Creating Thread of Executing using Runnable and using Sub-Class of Thread, Various Properties of Thread, using Synchronized Keyword, Using <i>wait</i> and <i>notify</i> , Collection Framework, Recap of Data Structures, The Collection Framework from java.util Package, Collection, Iterable, List, Set, Map, Queue, Dequeue, Utility Methods from Collections Class, Introduction to IDE	01	15 Hrs.

References:

1. Herbert Schildt, "Java, A beginners' guide", Oracle Press
2. Jain Pravin, "The class of Java", Pearson India
3. Horton I., "Beginning Java", 7th edition, Wrox India
4. Cay S Horstmann, Gary Cornell, "Core Java, Volume 1 – Fundamentals", Pearson Education
5. Ken Arnold, James Gosling, David Holmes, "The Java Programming Language", Addison-Wesley Pearson Education
6. Subramian, "Functional Programming in Java", VPragmatic Bookshelf
7. Urma R., Fusco M., Mycroft A., "Java 8 in action", Manning Publications

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Create appropriate classes using the Java Programming Language to solve a problem using Object Oriented Approach and use concepts of Functional programming
- Develop multi-user applications using Java Programming Language
- Create appropriate classes using Java Programming Language to solve a problem using Object Oriented approach and use concepts of functional programming
- Develop multi-threaded applications using the Java Programming Language
- Use data structures effectively to design applications in Java Programming Language

Course Name: Java Programming – Practicals

Course Code: DSC-C-CS-353P

Credits: 4

Course Outcomes:

The aim of this course is to enable students to

- **CO1:** Understand the concepts of Object-Oriented Programming Language and using Java
- **CO2:** Get good understanding of developing understanding of developing platform independent applications using java
- **CO3:** Get good understanding of multi-threaded applications using Java Programming language

Prerequisites:

Basic knowledge of C/C++ programming language

Contents:

Unit No.	Particulars	Credits	Time
1.	Unit Title: Basics of Java Programming Using Integrated Development Environment for Java Programming, Compiling & Executing Java Programs, Using Different Data Types in Java, Using various Operators & Performing Arithmetic Calculations, Decision Making with Java Programming, Looping, Implementation of Assertions, Use of Classes in Java, Constructors in Java, Exception Handling	01	30 Hrs.
2.	Unit Title: Inheritance, Interface and Commonly used Classes Implementation of Inheritance using Classes & Interfaces, Implementing Lambda, Implementation & Use of Packages in Java, Commonly used Classes & Methods	01	30 Hrs.
3.	Unit Title: File Handling, Collection Framework File Handling using Different Techniques like Input Output Streams: File.io, File.nio, Byte Streams and Buffered Input/Output, Reading, Writing and Creating Files; <i>Collection Framework: Interfaces for Collection:</i> Collection (Set, List, Queue, Deque) Map Interface	01	30 Hrs.
4.	Unit: Threads Processes and Threads, Implementing/Extending Thread Class & Runnable Interface in Java, Thread Objects, Pausing Execution, Join, Synchronization	01	30 Hrs.

References:

1. Herbert Schildt, "Java, A beginners' guide", Oracle Press
2. Jain Pravin, "The class of Java", Pearson India
3. Horton I., "Beginning Java", 7th edition, Wrox India
4. Cay S Horstmann, Gary Cornell, "Core Java, Volume 1 – Fundamentals", Pearson Education

5. Ken Arnold, James Gosling, David Holmes, "The Java Programming Language", Addison-Wesley Pearson Education
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Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Create appropriate Classes using Java Programming Language to solve a problem using Object Oriented approach and use concepts of functional programming
 - Develop multi-threaded applications using Java Programming Language
 - Use data structures effectively to design applications in Java Programming Language
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